PHILIPPINE ARMY MANUAL 6-0102

SIGNAL OPERATIONS

Proponent: Army Signal Regiment, Philippine Army
Promulgated: 15 April 2022

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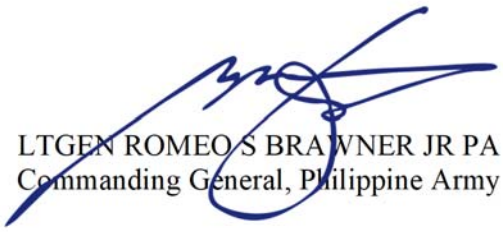
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Philippine Army Manual 6-0102 (PAM 6-0102) is promulgated on 15 April 2022 by authority of the Commanding General, Philippine Army



LTGEN ROMEO S BRAWNER JR PA
Commanding General, Philippine Army

Preface

1. The purpose of this manual is to prescribe principles, systems, and procedures for the signal operations. It is intended to help signal corps and signal assigned personnel to understand and implement PA communications systems utilizing different categories of signal communications. Also, this addresses the role and responsibilities of signal personnel and serves as a guide for signal planning during the military decision-making process.

2. This manual provides information and guidance for the Philippine Army personnel in planning, preparation, execution and assessment of their individual mission at all points along the signal support operational environments. This also guides commanders in performing command and control, and administrative functions to accomplish their mission.

3. It describes the broad framework to guide the management and employment of signal support operations. It covers the overview of signal operations to the Philippine Army, the roles, and responsibilities of Army signal organizations. Specifically, it includes information and communication technology (ICT) infrastructure, signal support operational environments; fundamentals of PA signal support; planning and execution; and sustainment of signal operations, which is critical to extend the signal support in every army operation.

4. The proponent of this manual is the Army Signal Regiment, Philippine Army. However, the Doctrine and Capability Integration Center (P), Training and Doctrine Command, Philippine Army is responsible in the review and updating of this publication. Send comments and recommendations to the Commander, TRADOC, PA, Camp O'Donnell, Sta. Lucia, Capas, Tarlac (Attn: Director, DACIC) or via email at dcannex@gmail.com and via website at doctrinecenterpa.com. Feedbacks to this publication are also welcome telephonically via the Bonifacio Telephone Exchange trunk line number 845-9555 or military local line number 7650 extension 4971.

5. Policies, rules, regulations and other publications inconsistent to this publication are hereby rescinded.

6. Unless this publication states otherwise, masculine nouns and pronouns do not refer exclusively to men.

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Chapter I
Philippine Army Signal Operations

0101. The Philippine Army (PA) signal operations (SO) serves as a linchpin to Army operations. It provides services that allow commanders to exercise command and control (C2) to their subordinate units. At the heart of the Philippine Army signal operations (PASO) are the units belonging to the Army signal corps (ASC), composed of highly trained signal officers and personnel dedicated to connecting Army units wherever they may be. Through the years ASC has undergone a series of organizational transformations, purposely to cope up with the ever-changing information, communication, and technological landscape.

The Army Signal Corps

0102. To better understand SO in the PA, we should first take a look at the rich history of the ASC. A history that is testament to the ASC's mettle, true to the adage that "the ASC does not claim to win wars, but the Army will not go to war without the ASC".

Communications Electronic

0103. The ASC traces its roots from the signal corps Armed Forces of the Philippines (AFP), which was created under the National Defense Act of 1936. It was tasked to establish signal communications in the ten military districts of the country. The signal corps saw action in 1941 when it fought side-by-side with the United States Army during the defense of Bataan and Corregidor against the Japanese Imperial Army, and later on during the guerilla resistance movement until the eventual liberation of the country in 1944.

0104. In 1954, AFP General Order Nr 455 activated the signal service group, which paved the way for the creation of the office of the chief, signal officer, the signal service battalion, the signal school, and the signal intelligence service. The Chief Signal Officer acts as the technical staff at General Headquarters (GHQ) and also served as the Commanding Officer of the signal service group.

0105. In March 1965, the signal service group was renamed as communications-electronics group or communication electronic (COMMEL) group and was tasked to provide, operate, and maintain the communications and electronics facilities which the AFP Chief of Staff exercised control.

0106. The COMMEL group was then re-organized into the communications-electronics service on 13 November 1978 as a reaction to its expanding roles and activities. The Army, as one of the major services of the AFP, in turn, reflected the organizational changes of the general headquarters with the office of the chief COMMEL service, PA as technical staff of the Commanding General.

Communication Electronics and Information Systems

0107. In keeping pace with the computer age and advances in information technology, on 01 May 1996 under HPA General Order Nr 701, the office of the chief COMMEL was elevated to a general staff status as the office of the assistant chief of staff for communications, electronics, and information systems or communication electronics and information systems (CEIS), G11. The Army signal group and the Army management information center were the two major Army units that fall under the functional supervision of the office. However, on 24 January 2005, under HPA General Order Nr 40, G11 was redesignated to G6 following a major reorganization in the staffs of the general headquarters AFP and the major services' headquarters.

Command and Control Communications and Cyber Systems

0108. On 13 April 2016, The Commanding General, PA approved the revised staff memo for the office of the G6, renaming it from the office of the assistant chief of staff for CEIS, G6, PA to the assistant chief of staff for command and control communications, and cyber systems (C4S), G6, PA in charge of formulating policies and programs for the development and use of information and communications technology (ICT) resources required for the effective command and control communications, and cyberspace operations of the Army.

Significant Organizational Changes

0109. One of the most significant organizational milestones in ASC history is the activation of the Army Signal Regiment (ASR) in 2011 and its eventual regularization in 2018. The creation of the ASR put in effect the following changes:

a. Professionalization of the signal AFPOS in the PA, carving a career path for signal officers and personnel

b. Reassignment of the territorial signal battalions from the infantry divisions to the ASR, as its organic unit, earned the signal corps the distinction of a specialized service

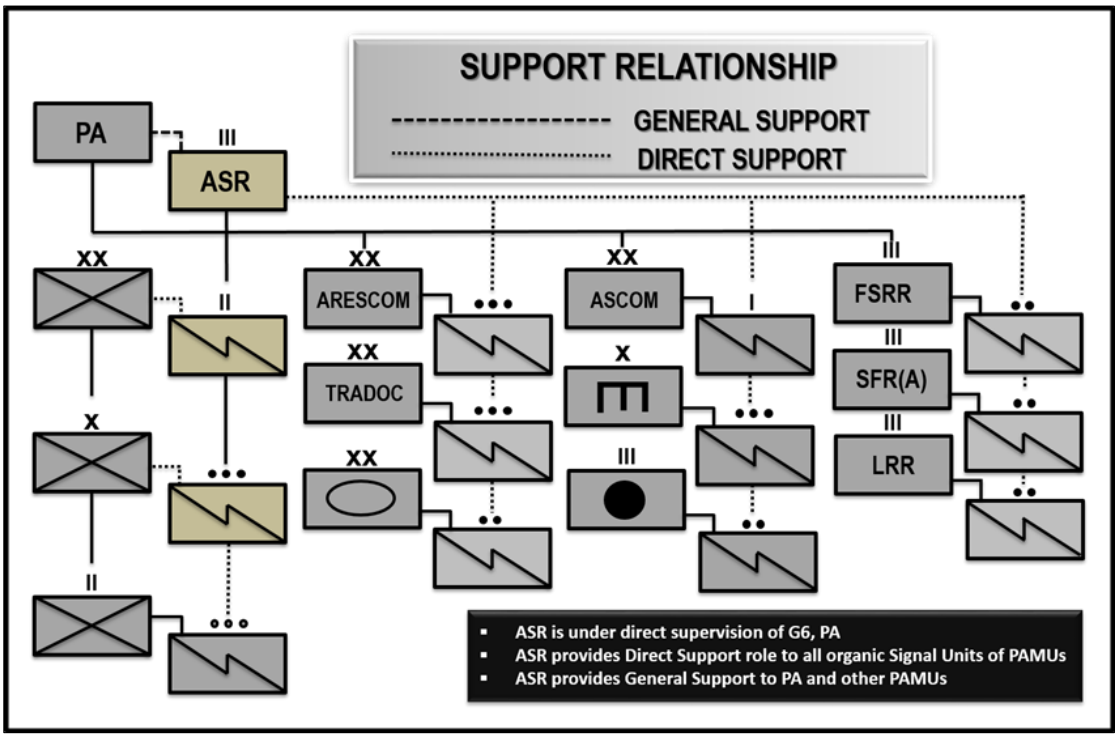


Figure 1.1 – Army Signal Regiment Support Relationship

0110. The creation of the ASR also gave way to the activation of two signal battalions, which will become the forefront of the PA cyberspace operations: The network enterprise technology battalion (NETBn) in 2018; and the cyber battalion (CyBn) in 2020. The NETBn is given the task to conduct information network operations (INO) mission while the CyBn is in charge of the defensive cyber operations (DCO) as well as limited active defense operations (ADO) mission. In addition, to secure and maintain the PA's freedom of action in the electromagnetic operational environment and deny the same for any adversary, activation of an electronic warfare battalion under the ASR looms in the horizon.

Impacts of Significant Events in Philippine Army History to Signal Operations

0111. In September 2013, rogue Moro National Liberation Front (MNLF) members stormed the city of Zamboanga. During the 19-day siege, the ASC was tasked to provide signal support to the Army and other major service components in a military operation in urban terrain (MOUT) environment. It also saw the advent of intelligence surveillance and reconnaissance (ISR) systems, used in pin-pointing enemy locations.

0112. In the aftermath of super typhoon 'Yolanda' in November 2013, commercial tele-communication services shut down, cutting-off in the province of Leyte from the rest of the country. Shortly after the onslaught of 'Yolanda', the Army's tactical communication was the first to break through to general headquarters and report on what had happened.

0113. In May 2017, ISIS-inspired local terrorist groups held a small city in Southern Philippines hostage, an event that would later on, be known infamously as the Marawi Siege. During this event, the AFP saw the proliferation of drones, automated weapon

systems, and precision missiles. It also gave rise to emerging C2 requirements such as digital maps, blue force tracking systems, live feed from drones, to include information dominance through social media. The Marawi Siege marked the PA's official entry into warfare in the digital age.

0114. In early 2020, the world was plagued with the dreaded corona virus, limiting human physical interaction and forever changing people's way of life. Internet technology has somehow mitigated the effects of the virus and even saved lives. The pandemic gave rise to people's dependence on social media, electronic messaging systems, and video-teleconferencing, at both personal and professional levels. The PA was not spared from the spread of the virus and had to come up with solutions to preserve the force using technology. Along with the Army's medical corps, the ASC was at the forefront in providing video-teleconferencing services as well as information systems development and automation.

The Signal Operations Operational Environment

0115. Like any military operations, the ASC units engaged in SO operate in a specific operational environment (OE). In the SO parlance, the OE is referred to as the area where the Army communication, electronics, and cyber systems are present and operate. The OE's environmental features, both natural and man-made, require significant consideration especially in the planning, preparation, execution, and sustainment of SO. Signal operates in three types of OE: physical; cyberspace; and electromagnetic spectrum.

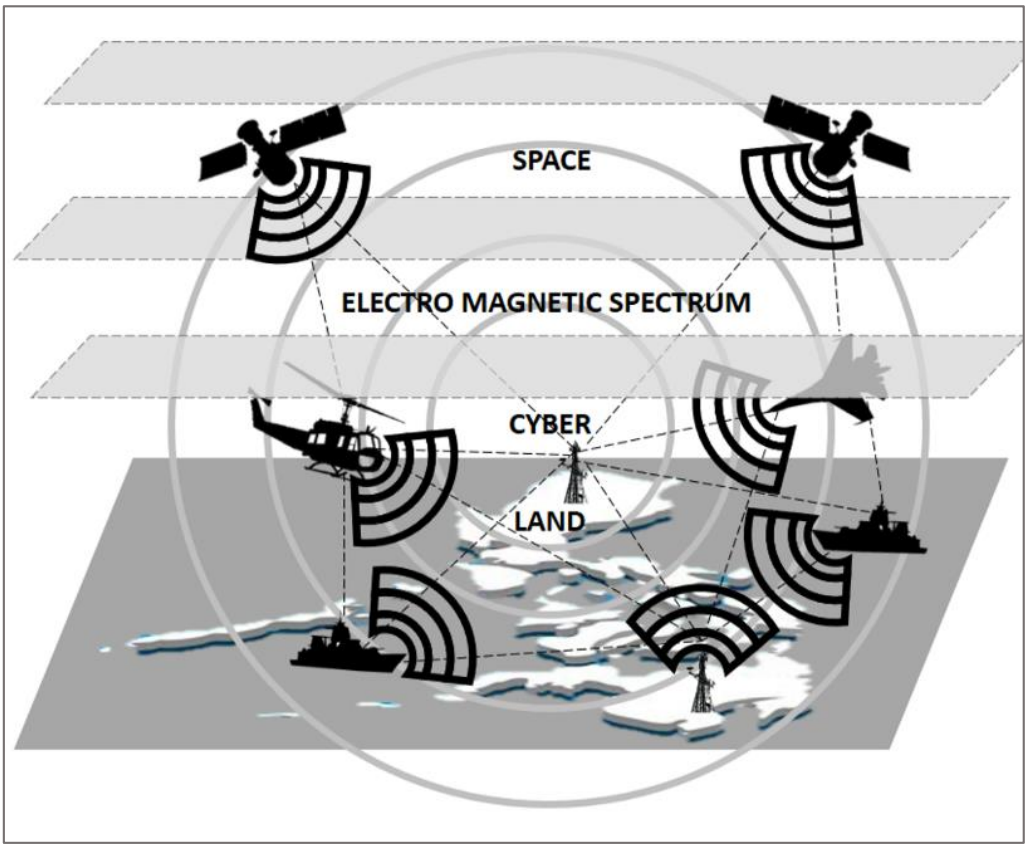


Figure 1.2 – Signal Support Operational Environment

Physical Environment

0116. The Philippines is an archipelagic country whose islands are separated by vast bodies of water. The distance between islands already poses a challenge in connecting Army units deployed all over the archipelago. The land locks are riddled with chains of mountain ranges and other natural features which affect the line of sight communication. Foliage and thickly vegetated forests affect signal propagation beyond the line of sight and even satellite communication. In built-up areas, especially in large cities, high-rise buildings, telecommunications towers, and facilities, coupled by high tension wires, affect radio frequencies and the electromagnetic spectrum. On top of this, the country serves as the path for several storms and typhoons formed every year, which often renders the Army's communication assets inoperable. Same consideration shall be applied when operating in foreign lands.

Cyberspace Environment

0117. It is the environment where there is PA cyber presence (Philippine Army, 2020). Cyber presence refers to hardware, software, and peopleware which make up the Army computer network infostructure. It is divided into three domains namely:

a. **Physical Domain**. This doesn't only refer to the land where the Army operates. In cyber, the physical domain includes the computer hardware, network devices, towers, and other transmission facilities.

b. **Logical Domain**. It refers to the configuration and relationship of cyber assets such as firewall configurations, network topology, and internet protocols.

c. **People Domain**. It refers to individuals, working either within or outside the organization, who interacts with the Army network infostructure through various cyber assets like computers, tablets, and cellular phones.

Electromagnetic Spectrum

0118. It refers to the range of frequencies of electromagnetic radiation present in the atmosphere. The frequencies range from lowest to highest, which accounts for all radio waves, infrared radiation, visible light, ultraviolet radiation, X-rays, and gamma rays. The electromagnetic OE facilitates the transmission of communications using tactical radio, microwave, cellular, and even wireless computer networks for both friendly and hostile forces. Control and dominance of the EMS is the holy grail of electronic warfare (EW).

Threats to Signal Operations

0119. Various incidents like Marawi and Zamboanga sieges placed the Signal Corps in a challenging situation as its capabilities were put into test by the following threats to signal operations. Most of these threats to signal operations may deliver a damaging effect to communication infrastructures and systems to targeted areas.

Information Warfare

0120. Refers to a threat's use of information activities, such as cyberspace attacks and EW, to gain an advantage in the information environment (Department of the Army, 2019).

Hybrid Threats

0121. It is the diverse and dynamic combination of regular forces, irregular forces, terrorists, or criminal elements acting in concert to achieve mutually benefitting effects (Department of the Army, 2019).

Threats in Cyberspace

0122. Threat actors can exploit system vulnerabilities and cause a loss of confidentiality, integrity, or availability of communications networks (Department of the Army, 2019).

Insider Threat

0123. Are hostile actors who intentionally compromise national security through deliberate actions (Department of the Army, 2019).

Army Signal Regiment

0124. The ASR is the PA's sole provider of signal support and services to address the threats listed above. Through its various transformation initiatives, ASR intends to provide secured network for the PA.

0125. One of the ASR's thrusts is to institutionalize the signal doctrine in collaboration with the Office of the Assistant Chief of Staff for Command and Control, Communications and Cyber Systems (C4S), G6, PA.

0126. It also continues to strengthen its organization, ensures the conduct of signal training of its troops, pursues the modernization program to acquire the latest ICT, and develop signal officers and personnel to ensure that the regiment has the right man for the right job.

0127. The regiment also acquired several ICT facilities and established networks and databases being maintained at secured locations that operate on a 24 hours basis daily.

0128. All these transformation efforts ensure that the ASR is equipped with the right technology and facilities and has the skilled personnel ready to perform the mission of the regiment for the PA.

0129. To fully understand the ASR organization and its capabilities, the organizational structure to include its subordinate units shown in figure 1.3.

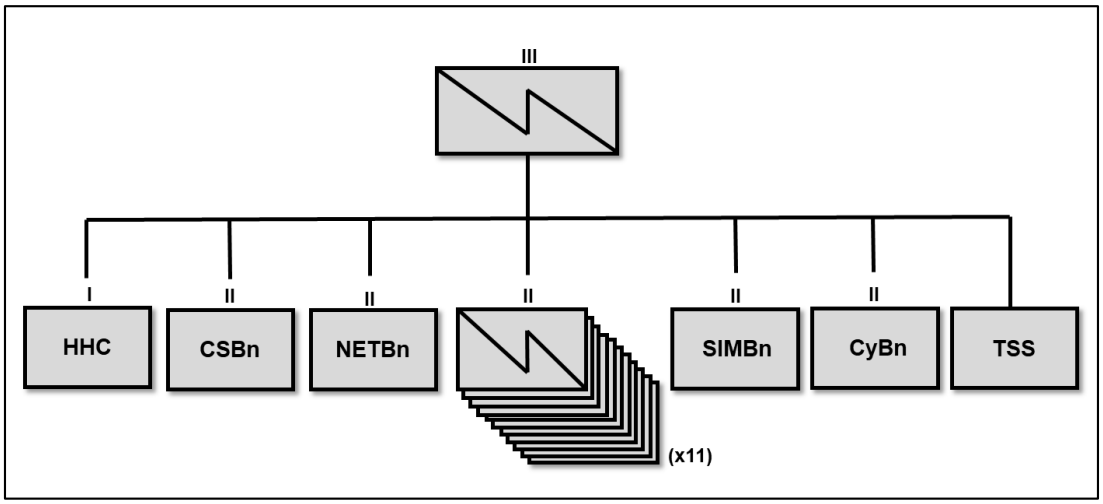


Figure 1.3 – ASR Organizational Structure

0130. The following are the ASR sub-units with their respective unit missions:

a. **Headquarters and Headquarters Company**. To provide command, control, and administration of the ASR.

b. **Command Signal Battalion (CSBn)**. To provide signal support to HPA post units, and other PA units in NCR.

c. **Network Enterprise Technology Battalion (NETBn)**. To operate, maintain, enhance, and secure the PA IT infrastructure in support of the mission of the ASR.

d. **Signal Battalion (SBn)**. To provide signal support to the Headquarters of the Infantry Division and Infantry Brigades.

e. **Signal Installation and Maintenance Battalion (SIMBn)**. To provide signal installation and maintenance support to the PA.

f. **Cyber Battalion (CyBn)**. To conduct cybersecurity operations to protect the PA cyber assets from cyber attacks across the PA domains of operations.

g. **The Signal School (TSS)**. To conduct specialized communications, electronics, and information systems education and training to personnel of the PA.

The Philippine Army Signal Support Services

0131. The PASO is a synchronized action of the ASC to connect Army units and affect the transmission, storage, and security of information within the PA infrastructure in support of Army operations. It is accomplished through various Signal Support Services which is divided into four categories:

a. Command and Control Communication

b. Cyberspace Operations and Electronic Warfare

c. Signal Repair and Maintenance

d. Signal Recovery and Restoration.

Command and Control Communication

0132. The main purpose of SO is to facilitate C2 by providing commanders the ability to communicate with their subordinate units. The US Army Field Manual 3.0 defines C2 as the "exercise of authority and direction by a properly designated commanding officer over assigned and attached forces in the accomplishment of a mission". Command and control communication (C3) is the employment of information and communication technology to facilitate the exercise of C2, in aid of Army operations. Signal support services under this category include fixed communication, tactical communication, and special purpose communication. While PAM 6-00 describes the categories of communication as fixed, mobile and special systems, tactical communications, described in this manual, is not a deviation to the foregoing categories, albeit it is complementary: tactical communication encompasses both mobile (on-the-move) and on-the-halt communications.

0133. **Fixed Communication**. The PA's fixed communication system (FCS) is a combination of wired and wireless technology, which serves as the backbone of the PA's voice and data communications. This includes the Philippine Army Bonifacio Telephone Exchange (BONTEX) and the Philippine Army Network (PANET) respectively. The BONTEX has long since served the PA, connecting the PAMU telephone systems with the Headquarters Philippine Army (HPA) and even up to GHQ AFP. With the advent of voice-over IP technology, the Army is slowly shifting to IP-based telephony using PANET. The PANET is the Army's secured and managed intranet services where PA's administrative and mission-critical information systems are accessed. At present, the PA's fixed communications are extended down to the Army Brigades through the deployable brigade tactical operations center system (DBTOCS). Services under fixed communication include telephone services; video-teleconferencing, electronic mail and collaboration systems, and information systems under the Army operations domain.

0134. **Tactical Communication**. Despite the proliferation of computer networks and satellite communication, voice communication remains to be the bedrock of tactical communication (Blumberg, 2020). Tactical communication refers to military-grade on-the-move (OTM) and on-the-halt (OTH) communication devices operating under the radio frequency spectrum. It is composed of tactical radios and tactical radio networks

(Department of the Army, 2020). Technologies under tactical radios include high frequency (HF), very high frequency (VHF), and ultra-high frequency (UHF) radios. Tactical radio network, on the other hand, is comprised of the PA combat net radio (CNR) system using radio over IP (ROIP) and ad-hoc network technologies. Non-military specification radios are used to complement tactical radios primarily as a means to establish communication with other agencies especially during times of emergencies or disasters.

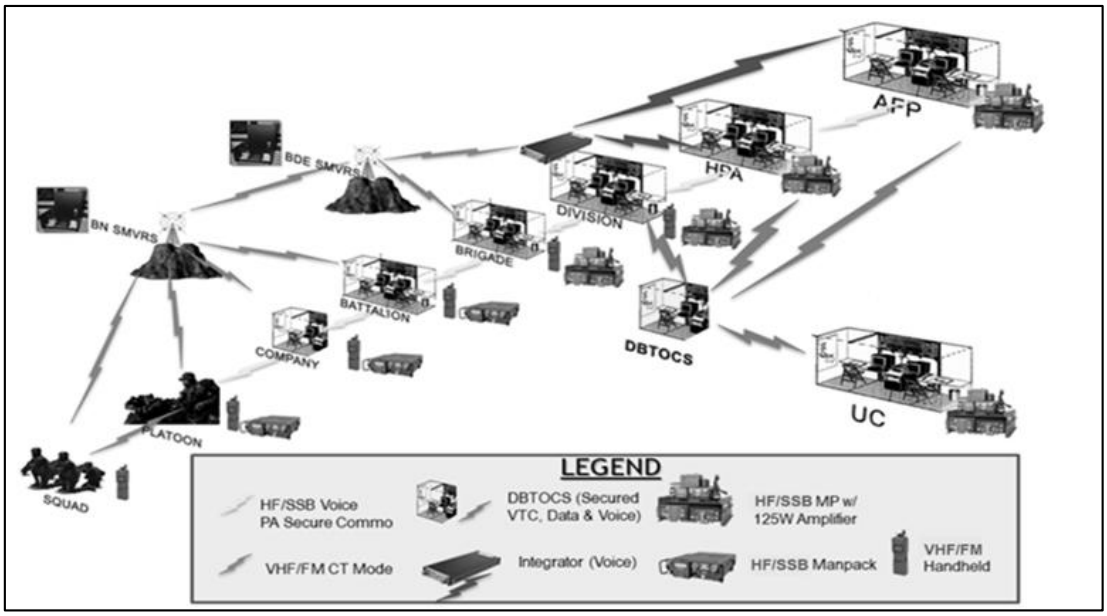


Figure 1.4 – Combat Net Radio System

0135. **Special Purpose Communication (SPC)**. To augment the PA's fixed and tactical communications, the Army uses commercial devices for routine or administrative communication. This service falls under SPC. The SPC includes commercial videoconferencing platforms, cellular phones, and satellite phones. Technologies under SPC are utilized with utmost prudence, guided by existing communication security (COMSEC) policies.

Cyberspace Operations and Electronic Warfare

0136. Cyberspace operations is defined as the employment of cyberspace capabilities where the primary purpose is to achieve objectives in or through cyberspace (Joint Staff, 2018). PA Cyberspace Operations has three distinct missions:

a. Information Network Operations

b. Defensive Cyber Operations

c. Active Defense Operations

0137. Out of the three missions, the ASC is responsible for two, INO and DCO (Philippine Army, 2020). ADO, on the other hand, will be a collaborative effort between the ASC, Army intelligence, and civil-military operations (CMO) units.

0138. Electronic warfare consists of actions that use electromagnetic energy to destroy, neutralize or reduce enemy combat capability. It takes advantage of the use of the electromagnetic spectrum by the opponent while ensuring efficient use of its electromagnetic emissions (Victoria & Christianity, 2019). The EW is divided into three broad objectives: support; attack; and protect.

Signal Repair and Maintenance

0139. With very meager resources, the ASR needs to come up with a maintenance strategy that would see to the readiness of communication equipment and ensure continuity of service to the operating troops. There are three main mechanisms for signal repair and maintenance: The signal maintenance and training facility (SMTF); the signal mobile maintenance and repair team (SMMART); and the DND-AFP electro-optic repair and maintenance facility.

0140. SMTF are facilities that are equipped to conduct levels two to four radio repairs. To be more responsive to the maintenance requirements of the operating troops, SMTFs are built in strategic locations, closer to the line units.

0141. The ASR created SMMART to be more responsive to maintenance requirements of subordinate units. Said teams regularly visit the brigades, battalions, and even down to the company level to inspect C4S equipment; conduct level one and two repairs for tactical radios, night fighting systems (NFS), and computers; and conduct C4S related refresher training.

0142. The DND-AFP electro-optic repair and maintenance facility is a facility dedicated to the repair of NFS. Although it is operated and maintained by the SIMBn, this facility caters to NFS repair requirements of other major services.

Signal Recovery and Restoration

0143. Learning from the lessons of Super Typhoon Yolanda in 2013, communication equipment and facilities are not impervious to natural calamities. Whether it is a natural or manmade calamity, the ASC needs to be prepared to repair and restore damaged communication facilities to ensure continuity of service. To do this the ASR organized deployable signal teams ready to answer the call. Deployable signal teams (DST) are categorized into four: tactical operations center (TOC) DST; C3 restoration teams; network restoration teams; and cyber security incident response teams (CSIRT).

The Philippine Army Signal Operations Framework

0144. The PA signal operations is anchored on connect, sustain and recover (CSR) methodology: Connect ensures connectivity of all Army units; Sustain ensures that the connectivity is maintained throughout the operations, and recover is the ability to effectively repair and restore communication if it is attacked or destroyed. The CSR methodology is practiced in conjunction with the conduct of military operations cutting across the different operational environments.

0145. The SO framework depicts the application of the different SO services to the operational environment using various types of communication technologies in support of the Army operations. The SO addresses the various threats that may hamper the delivery of these services and the success of the operations by using the CSR methodology cutting across the different operational environments.

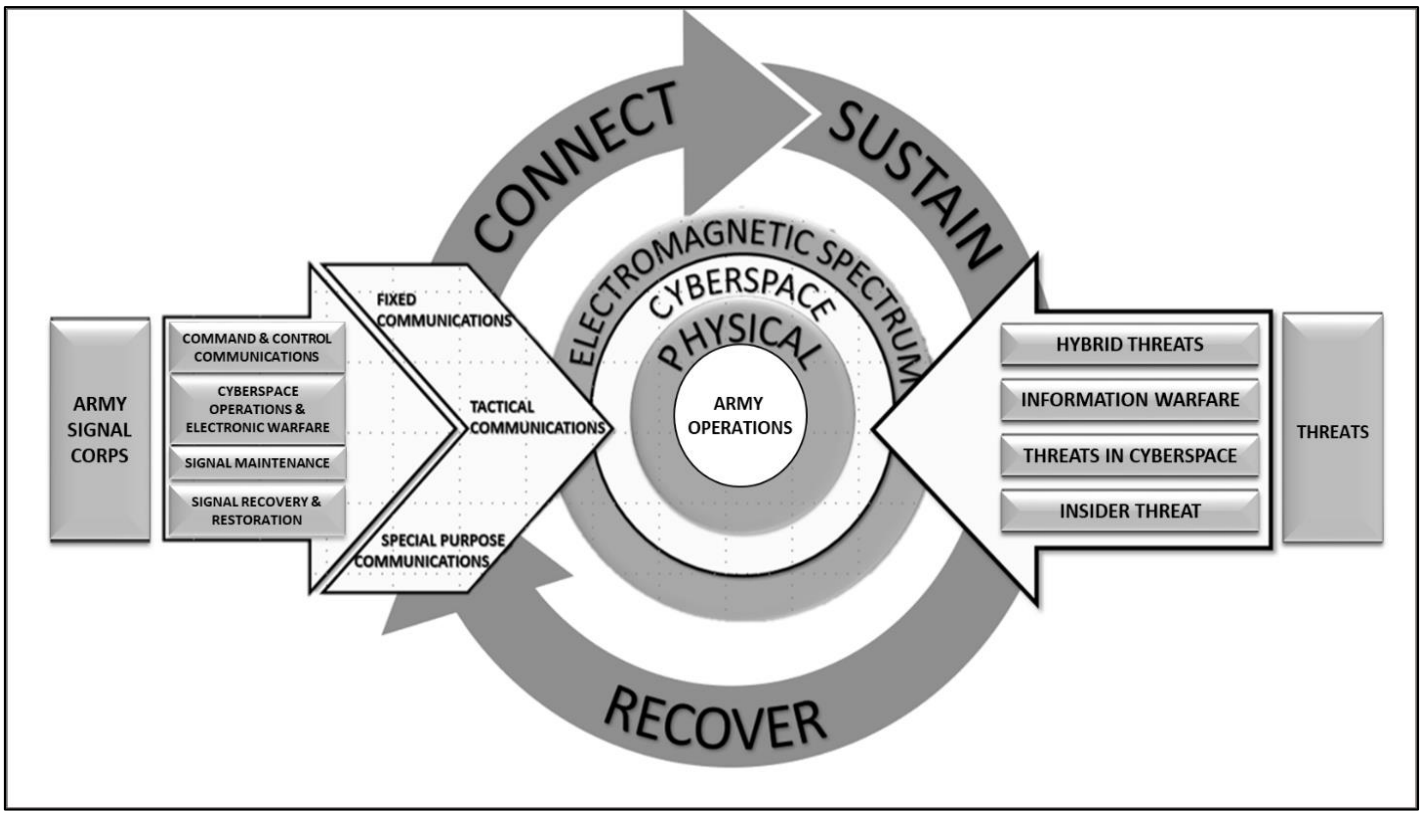


Figure 1.5 – Signal Operation Framework

Chapter II
Fundamentals of Philippine Army Signal Operations

0201. The main purpose of PASO is to provide communication services in support of combat operations. Success in the battlefield depends on the timely receipt of accurate information presented in a format that can be digested readily by the commander and staff (Frater & Ryan, 2015). PASO affects the transmission, storage, and security of information between commanders and their subordinate units across the operational environment. It is guided by existing operational concepts and strategies and implemented using fundamental principles in PASO.

Methods and Strategies

0202. Signal and C4S. As the ASC's focus transition from signal to C4S, the publication of PAM 6.0 describes the distinction between the two concepts. Signal is the action that supports Army operations. On the other hand, C4S consists of capabilities, enabling signal to support Army operations (Philippine Army, 2017).

0203. Technological advancement and the future of warfighting. The PA operating concept 2021 described how technological advancement has exponentially changed the operational landscape. The employment of non-kinetic capabilities such as cyber can now affect, not only the weakening of communication capabilities but even cause damage to lives and properties.

0204. Cyberspace. GHQ, Letter Directive number 28 prescribes the adoption of Cyberspace as the fourth domain of AFP operations, alongside the Land, Sea, and Air.

0205. Cyberspace Operations. The PAM 6-0102-04 defines cyberspace operations as the employment of cyberspace capabilities, where the primary purpose is to achieve objectives in or through cyberspace.

0206. Electronic Warfare. Based on the PA capability assessment 2021, emerging threats in both local and international environment has given rise to the need to develop EW capabilities.

0207. Vigilance Net. GHQ, SOP number 7 dated 26 June 2001, prescribes the use of a set of communication procedures between units of the AFP, during operational exigencies.

Fundamental Principles of Signal Operations

0208. Conducting signal operation requires a disparate set of knowledge and skill depending the signal support services being provided. While knowledge and skills requirement are different for each services, the guiding signal principles for which these services are being provided transcends the different categories of signal support services.

Operations Focused

0209. Communication is an enabling capability that supports combat operations. Counterinsurgency operations, however, entails a prolonged period of combat operations which often takes its toll on resources and equipment like communication devices. Worst case scenario would be, if combat operations are stopped or put on hold due to discharged battery packs or the unit has lost contact with its mother unit. Communication support should never be the limiting factor to the commander's ability to execute a mission (Department of the Army, 2019).

Interoperable

0210. PASO requires that all units, within the Army or with other major services, would be able to communicate with one another in the course of the combat operations. Interoperability requires compatibility and standardization among communications and automated information systems (Department of the Army, 2019).

Redundant

0211. Communication devices don't often work the way we want them to. A good practice must be to set up communication redundancy strategies to ensure continued communication services. Redundancy provides multiple transmission paths, backups, self-healing strategies, and data replication (Department of the Army, 2019).

Scalable

0212. Combat operations may often escalate faster than the Army must anticipate. Signal support should have the ability to scale and expand its services based on the prevailing situation. Signal forces must provide a network that is tailorable to adapt, based on phases of an operations, to enable command and control at home station, en route, and in deployed environments (Department of the Army, 2019).

Secured

0213. Communication technologies must have advanced through the years, yet the concept of security will always be the guiding principle in transmitting the information. Regardless of the type of communication services that are being provided, provisions under the COMSEC policy still govern.

Signal Operations Activities

0214. In each of the four, Signal Support services are distinct activities that are practiced to ensure success in PASO. These activities require different communication technologies, as well as the skills and competencies of every signal officers and personnel.

Command and Control Communications.

0215. Command and control communication (C3) is the employment of information and communication technology to facilitate the exercise of C2, in aid of Army operations.

0216. **Signal Estimate**. Like any military operations, signal officers are responsible to have a clear understanding of the type of environment the Army units will be operating in and its effect on various signal equipment. The Signal Estimate sets the stage for signal support operations. It allows signal operators to determine the appropriate signal support strategy that is going to be applied to the operations. Signal estimate is as dynamic as the combat operations itself and may change from time to time.

0217. **Signal Planning**. Signal plans are the cornerstone of every signal support operations. Its success relies heavily on the quality of signal estimates conducted before the actual planning. The signal plan outlines the signal challenges in a given operational environment as well as the technological solution that may address these challenges. Significant to the plan is the sustainment and recovery strategy to sustain the Army operation.

0218. **Electromagnetic Spectrum Management**. SO oftentimes entails connecting a myriad of units performing in different capacities and command relationships, using disparate communication technologies, and operating in various frequencies. Spectrum management is the planning, coordinating, and managing the use of the electromagnetic spectrum through operational, engineering, and administrative procedures. This activity should be conducted in congruence to existing laws and policies of the country, set forth by the national telecommunications commission (NTC).

0219. **Support to Command and Control**: 1) Operations Center Support. All data and information being collected by sensors and communication devices are made available to the commander in a single facility known as the operations center. It serves as the nerve center where the signal support services converge. 2) Mobile Command Center (MCC) Support. Operations center facilities should not be confined to fixed facilities. The commander should be given the capability to exercise command and control functions wherever he may be. MCCs provide the commander the ability to establish his operations center wherever he chooses. It is equipped with the necessary C4S capabilities needed by the commander to lead.

0220. **Other Signal Activities**: 1) Night Fighting Systems Support. Signal was given the task to provide NFS capabilities to PA units. It provides support through training as well as repairs and maintenance. 2) Public Address System (PAS) Support. Another ancillary duty of the ASC is to provide PAS support, especially during command

activities. 3) Research and Development. Part of signal units and offices' functions is to pursue innovative concepts that would help improve systems and enhance signal capabilities to accomplish the Army's mission.

Cyberspace Operations and Electronic Warfare

0221. Activities in Cyberspace Operations and Electronic Warfare is focused in attaining information dominance. It is geared towards the protection of the PA's information infrastructure, while trying to deny the enemy's ability to use cyberspace and the electromagnetic spectrum against the friendly forces.

0222. **Cyberspace Operations**. 1) Information Network Operations. The purpose of INO is to build the PANET and effect the expedient restoration of connectivity and services after a computer emergency. Part of the INO is information system development and integration. 2) Defensive Cyberspace Operations. The purpose of DCO is to defend PANET from all forms of cyber threats and attacks. It also provides a prompt and effective response to cyber incidents. 3) Active Defense Operations. The purpose of ADO is to exploit the enemy's cyber vulnerabilities to defeat and deny them the ability to use cyberspace against Army forces. (Philippine Army, 2020)

0223. **Electronic Warfare**. 1) Support. The purpose of this EW measure is to sense Electromagnetic (EM) activities. These are Informative activities and cyber measures to detect threats (Thales, n.d.). 2) Attack. This EW measure comprises defensive activities and cyber measures to neutralize threats (Thales, n.d.). 3) Protect. This EW measure prevents threatening EM activity (Thales, n.d.). These activities may range from frequency hopping, anti-jamming, and counter-countermeasures.

Signal Repair and Maintenance

0224. With very meager resources, the ASR needs to come up with a maintenance strategy that would see to the readiness of communication equipment and ensure continuity of service to the operating troops.

0225. **Repair Facility Operation**. The PA maintains two types of signal repair facilities: the SMTF; and the DND-AFP electro-optic repair and maintenance facility. Both types are equipped to perform high-level repairs and maintenance for both tactical radios and NFS respectively. Access to the SMTF, in particular, is brought closer to the line units by establishing extension facilities at western and northern Mindanao where there is a predominance of troops.

0226. **Field Signal Support**. The creation of the SMMART allows the ASC to be more responsive to the repair and maintenance requirements of the line units without the need of bringing the equipment to the SMTF. At the moment territorial signal battalions perform the mobile repair and maintenance, by visiting units down to the battalion level. It is envisioned however that through SMMART, signal operators in the field will develop a level of competency that would allow them to perform field-level repairs. The new field support concept for network and mission command systems embraces Soldiers as the first line of defense for troubleshooting (U. S. Army, 2014).

Signal Recovery and Restoration

0227. Activities under Signal Recovery and Restoration ensures the readiness and resiliency of all signal assets in order to deliver effective and unhampered signal support services to Army operation.

0228. **TOC DST**. It is primarily organized to restore connectivity of command centers damaged by natural and manmade calamities. The following are the specific functions of a TOC DST:

a. Creating and maintaining a TOC deployment plan

b. Operationalize the Mobile Command Center

c. Establish communication with subordinate units

0229. **C3 Restoration Teams**. Its general function is to repair broken communication systems caused by disasters. The following are the specific functions of a C3 Restoration Team:

a. Creating and maintaining a C3 Restoration plan

b. Investigating and analyzing incidents

c. Communicating with officers and personnel about incidents as needed

d. Conduct repair and restoration

0230. **Network Recovery Teams**. Its primary task is to restore PA network connectivity. The following are the specific functions of a network recovery team:

a. Creating and maintaining a network recovery plan

b. Investigating and analyzing incidents

c. Communicating with officers and personnel about incidents as needed

d. Conduct repair and recovery

0231. **Cyber Security Incident Response Team (Matthews, 2018)**. The team's primary task is to address catastrophic cyber attacks. The following are the specific functions of a CSIRT:

a. Creating and maintaining an incident response plan (IRP)

b. Investigating and analyzing incidents

c. Managing internal communications and updates during or immediately after incidents

d. Communicating with employees, shareholders, customers, and the press about incidents as needed

e. Remediating incidents

f. Recommending technology, policy, governance, and training changes after security incidents

Chapter III
Planning and Execution

0301. Planning is significant to be able to organize resources and activities efficiently and effectively to achieve the desired goals. Signal planners must be committed to handling critical tasks to accomplish signal support operations mission. Likewise, signal planners must be subject matter experts on signal functions, C4S capabilities, and signal support operations.

Signal Estimate

0302. Signal estimate is an orderly and logical examination of all factors, known or assumed, that may affect the successful accomplishment of signal support operations mission. It is formed by developing separate estimates for each signal support and service, and C4S capabilities available or a single estimate that consider all means required for signal support operations. It is revised as the situation changes and new facts are introduced.

0303. The keyword "METAL" aids the Signal planner in the preparation of the signal estimate in any given situation and location. This keyword takes into account the "Mission" in precise terms, the "Enemy" signal capabilities, the "Troops" available to accomplish the mission, the "Area of Operations", the "Logistical" requirements to assure the successful accomplishment of the mission. Signal estimate is also referred to as C4S Estimate.

The Signal Planning Process

0304. To ensure that all aspects of signal support operations are systematically addressed, a series of logical steps has been established called the signal planning process. The signal planning process is in conjunction with the military decision-making process (MDMP) and troop leading procedure (TLP) for small units. Detailed planning may differ from each of the PA signal support and services, but all apply the fundamental principles of signal support operations.

Develop Signal and Command & Control Communication and Cyber System Requirements

0305. The signal planner obtains and evaluates all sources of information about the signal support and services, and C4S capabilities required by the command or unit. These are the sources:

a. **Commander's Guidance or intent**. In the form of the concept of operations provides ample sources of information. As different situations call for varying approaches, it also needs different signal support and services, and C4S capabilities to be employed to achieve the desired commander's intent. Signal planner provides different course of action, however, the commander

specifically directs the means to be provided as well as the priorities of the establishment.

b. **Staff Guidance**. The signal planner also derives information and guidance from other members of the staff. Since these personnel are the users of the signal support, services, and C4S capabilities to be employed.

c. **Guidance from Higher Headquarters, Subordinate and Adjacent Units**. The signal planner determines other signal and C4S requirements by securing guidance from higher, subordinate, and adjacent units. This will provide the right information as to the specific requirement and support of higher headquarters. Likewise, it will provide detailed information on the expected signal support and services needed by subordinate and adjacent units.

d. **Standard Operating Procedures**. A comprehensive standard operating procedure (SOP) will minimize the time and effort required in planning. Thus, the signal planner obtains guidance and information on the different C4S capabilities installed, operated, and maintained from published SOP.

e. **Communication-Electronics Operating Instructions (CEOI) and Communication-Electronics Standing Instructions (CESI)**. The CESI, contains information explaining the use of items found in the CEOI, "How to do". The CEOI provides specific instructions for the operation of C4 Systems, "What to do". From these two documents, the Signal planner obtains information on what to employ in any given situation.

Determine Available Resources

0306. The signal planner determines all available C4S resources of the organization to provide signal support and services to accomplish the mission. To facilitate the determination of available resources, the Signal planner maintains an up-to-date chart that shows available C4S equipment, personnel, and funds.

0307. **Equipment**. In planning, consider C4S equipment and maintenance readiness based on the authorized table of equipment (TE), on-hand, in-use or serviceable, non-mission capable or unserviceable, available or categorized as reserve pool and capabilities of every equipment and systems.

0308. **Personnel**. In planning, consider signal personnel readiness based on the authorized table of organization (TO), actual fill-up, the status of training, critical shortage, committed or deployed, and reserve.

0309. **Funds**. In planning, consider the needed fund support for the installation, operations, and maintenance of C4 systems and other administrative requirements in conducting signal support operations.

Develop Priorities

0310. In this step, the signal planner compares and evaluates the requirements to the capabilities. This is to determine the requirements that can be supported. If adequate

resources are not available, the Signal planner determines the priority for the signal support and services to be rendered. These priorities are determined in terms of signal supportability and its importance to the accomplishment of the assigned mission.

Develop Recovery Procedures

0311. While plans are made to ensure the success of the operation, signal planners must include in their plan, provisions for other contingencies in the event of:

a. **Interrupted Connectivity**. Communication interruption may be in the form of configuration issues or damaged communication equipment. The signal plan must include: redundant communication; equipment repair and replacement; and on sight technical support.

b. **Electronic Warfare**. Tactical communications are susceptible to jamming and interception. The signal plan must include countermeasures as well as counter-countermeasures to support and protect the operating units from electronic attacks.

c. **Cyberattacks**. Cyber threats are prevalent in the cyberspace domain. As the PA faces a more and more information centric warfare, cyberattacks against the Army network is imminent. The plan must include provisions for: monitoring the PA network to include the mobile ad-hoc network; and procedures for CSIRT and first responder employment.

d. **Compromised Communication Frequencies**. In the course of the operations communication equipment may fall into the hands of the adversaries, compromising operational radio frequencies. The plan must include procedures for changing frequencies in the most efficient and effective means and in the shortest time possible.

Prepare a Tentative Plan

0312. This step is the most complex and varies depending on the specific C4S capabilities and signal support and services being planned. In general, planning the entire signal support operations includes the following:

a. Planning the general concept of signal support and services includes primary, alternate, contingency, and emergency plans

b. Allocation of resources to the requirement

c. Installation of C4S centers and other facilities must be based on factors such as requirement density, tactical disposition of units, location of support units, terrain, propagation condition, projected operations, accessibility, administrative and logistic requirements

d. Security measures

e. A general plan of interconnections and alternate routing

Finalize the Plan

0313. In this step, the Signal planner completes the detailed planning and checks the plan in its entirety including the details of various C4S capabilities and signal support and services being planned. The signal planner ensures the following:

a. The plan accomplishes the signal support operations requirements of the command or units

b. The plan is practical and that the resources and capabilities are adequate for executing the plan

c. The emphasis on the plan agrees with the priorities that had been determined

d. The areas of responsibilities for the different C4S centers and facilities that will be established

e. A survey or reconnaissance and propagation of the site was made

f. The locations of the different C4S centers and facilities were pinpointed during the survey or reconnaissance

g. Signal coverage and gaps were determined during the propagation

h. Security measures are employed within the operational environment

Publish the Plans and Orders

0314. In this step, the plans and orders with the required diagrams and instructions are prepared and published. The signal planner recommends to the operations officer inclusion of items for recommendation/decision in the entirety of the unit's plans and orders in the form of a signal estimate. Also, prepares the signal plans and orders or the C4S annex to the operations order/plan as well as the appendices to the annex and tabs to the appendix as required.

Signal Support to Joint Operations

0315. The success of signal support in joint operations is dependent on good communication interoperability among major services following established procedures and protocols as enumerated below.

Vigilance Net

0316. The use of the vigilance net is reserved for emergency and exigent situations. As a practice, however, a vigilance net is also employed during joint operations. Since this method operates in a less secured mode, it is susceptible to attack and eavesdropping. To mitigate the vulnerability of operating in the vigilance net, ciphertext, and double talk should be practiced.

Lending Equipment to the Other Service

0317. Another strategy that may be adapted during the conduct of signal support in joint operations is for the Army to provide or lend communication equipment to other services. This equipment will serve as the conduit for the other service to communicate with the Army operating troops.

Embedding Army Radio Operators to Other Service

0318. During joint operations the Army may embed radio operators to the other services. These operators act as a relay between the operating units and the assets provided by the other major services.

Embedding Radio Operators from Other Service to the Army

0319. The most practiced strategy in SSO during joint operation is to embed radio operators from the other service with the tactical operations center that is closest to the operating units and where the combat commander or decision-makers are. It simplifies the operational radio net and facilitates effective command and control, considering that the combat commander has direct contact with the assets.

Signal Support to Coalition Operations

0320. Coalition-type operations are military operations where the PA operates with units from other nations. This type of operations may be in a form of military exercises, military assistance in disaster response operations, or peace-keeping operations.

0321. Peculiar to coalition type operations is maintaining a sovereign network for each nation. Critical to this type of operations is the identification of points of integration where each unit in the coalition may communicate with one another.

0322. SSO strategies used in joint operations may be applied to this type of operations

Signal Support during Humanitarian Assistance and Disaster Response Operations

0323. Army units engaged in humanitarian assistance and disaster response (HADR) operations are guided by the provisions of the national disaster response and risk mitigation (NDRRM) Plan and the AFP disaster response plan. The primary mechanism for disaster response is the incident command system (ICS) wherein the military has been given a specific role. Unlike the regular military operations, signal support is not only intended for military units engaged in HADR operations. Support is also extended to other stakeholders involved in the response.

0324. The radio net developed during HADR operations must be different and separate from the Army's secured operational net. The main purpose for the PASO during HADR is to connect various entities engaged in the operations using various communication equipment. As such, HADR radio nets shall be considered unclassified and unsecured.

0325. Deployment of Deployable Signal Teams. In the event of a disaster whether anticipated or sudden onset, DSTs are activated and put on standby for deployment. The DST's main purpose is to restore command and control and restore or repair lines of communication when needed.

0326. Deployment of Mobile Command Centers. Aside from the DSTs the ASR also prepares to operationalize the mobile command center. This is especially helpful during the sudden onset of disasters when establishing communication with line units, at the soonest time possible, is of utmost importance.

Chapter IV
Sustaining Signal Operations

0401. Sustaining military operations is extremely important. The PA signal support mission is developed for a specific purpose and with the capability to integrate C4S resources into the unit. To support current and future operations, existing resources must be carefully maintained and made available.

0402. Successful signal operations depend heavily on the standard of signal personnel operating the equipment. It follows the reliability of its components. In particular, the hardware is commonly referred to as a computer or equipment, the software, a component that operates a programmed system-based machine; and 'Peopleware,' considered to be the most critical and significant of the three.

Equipment Maintenance

0403. With the rapid development of ICT worldwide, the PA C4S plans and programs will supplement the attainment of Army Transformation Roadmap (ATR) 2028 by sustaining the serviceability of on-hand C4S equipment and facilities, and upgrading them through the projects under the PA capability upgrade program (CUP), strategic initiatives of ATR 2028, and allocations from available capital outlay (CO) funds.

0404. The increasing number of C4S equipment in the PA inventory will also mean an increase in its maintenance. The on-hand C4S equipment and existing C4S facilities of PA are at risk of deterioration once non-observance of prescribed repair and maintenance procedures by PA units. In some instances, negligence on the part of the individual personnel operating this C4S equipment and facilities led to its eventual breakdown.

0405. Repair and maintenance of on-hand C4S equipment and facilities in PA inventory require concerted efforts of the individual personnel/operator, all levels of PA leadership, using/recipient and signal units as a whole, as well as, the support of partner C4S equipment suppliers. These entities shall be required to follow set guidelines and specific procedures to be observed by PA units to establish reliable and responsive signal support activities to sustain the serviceability of issued C4S equipment and facilities.

0406. The extension of the operational and service life of existing C4S equipment and facilities must therefore be given equal importance to PA. This can be achieved through aggressive repair and maintenance programs, the appropriation of c4s maintenance and other operating expenditures (MOOE), and the dedication of PA personnel to maintain the serviceability of C4S equipment and facilities in the PA inventory.

Signal Maintenance Units

0407. To sustain the signal capability of the PA, the following signal units are responsible for specialized ICT repair and maintenance at three command levels.

a. Signal Installation and Maintenance Battalion, ASR. (Strategic Level) with the following capabilities: 1) Provide repair, installation, and maintenance of the C4S of the Philippine Army. 2) Provide level 3 and 4 repairs of C4S equipment of the Philippine Army. 3) Conduct test, evaluation, research, development, and special studies on C4S equipment for the Philippine Army.

b. Signal Installation and Maintenance Company (SIMCoy), Signal Battalion, ASR. (Operational Level) with the following capabilities: 1) Plan, organize, direct and supervise repair and maintenance of all C4S equipment of the PAMUs. 2) Provide levels 1 and 2 repairs and maintenance of C4S equipment of the PAMUs. 3) Conduct tests, evaluation, research, and special studies on C4S equipment for the PAMUs.

c. SMMART, Signal Battalion, ASR – (Tactical Level) with the following capabilities: 1) Provide level 1 and 2 repairs and maintenance of C4S equipment to increase maintenance readiness of the PAMU subordinate units. 2) Provide consultation services to improve the signal operations and maintenance readiness of PAMU subordinate units. 3) Conduct on-site signal training to PA personnel performing signal functions at respective units.

0408. In support of major battles and national emergencies, the FSMMART will be co-located at the tactical command post (TCP) of the command group, or as directed by the ground command, and will serve as a forward signal maintenance station to address the need to sustain signal support for frontline units.

Levels of Maintenance and Responsibility

0409. Shown in Table 1 are the institutionalized standard ICT repair and maintenance levels to be observed with corresponding unit/individual responsible to include a description of activities involved:

Table 4.1 – ICT Repair and Maintenance Level

Level	Responsible Unit/Person	Description/Activities
1	Radio Operator and/or User/Client	- Visual inspection of external parts - No Maintenance Repair Kits involved - No Tools/Test Equipment involved - Use of Operator's Manual - Requires Operator's Training - No spares involved
2	S IMCoy, Signal Battalion, ASR	- Evaluation of physical relationship of parts - Use of Basic Tools/Test Equipment - Requires Systems Maintenance Training Use of Radio Programming Application

		- Includes assembly/disassembly of Accessories
3	SIMBn, ASR	- Opening of radio casing and removal of modular parts for evaluation/replacement - Use of Special Tools/Test Equipment - Use of Unit Maintenance Manual - Requires Systems Maintenance Training and experience in handling modules - Use of Software (RPA, CPA, etc.) - Handling of modular parts
4	SIMBn, ASR	- Opening/Repair/Replacement of parts of the modules (Transistors, Capacitors, - Integrated Circuit, Resistors, Diodes etc.) - Use of Automated Maintenance Repair Kits - Use of Special Tools/Computer-Aided Test Equipment - Requires Systems Maintenance Training Level IV Skills - Use of Systems Software (RPA,CPA,etc.) - Handling of spare parts for modular parts

0410. In addition, suppliers of C4S equipment must provide level 3 and 4 repairs and/or replacement to unserviceable PA-procured C4S equipment under the service warranty period.

Repair and Maintenance Procedures

0411. Repair and maintenance of on-hand ICT equipment and facilities in PA inventory require the concerted efforts of the individual personnel/operator, all levels of PA leadership, recipients, and signal units as a whole, as well as, the support of the partner ICT equipment suppliers.

0412. Users of ICT equipment are required to follow the specific guidelines and procedures needed by the PA units to establish reliable and responsive signal support activities to maintain the serviceability of ICT equipment and facilities provided. The procedures, therefore, include the following maintenance efforts of PA to extend the duration of each facility beyond its service life.

Preventive Maintenance

0413. This includes preventive maintenance checks and services to detect early signs of equipment failure. This maintenance normally consists of inspection, cleaning, preserving, lubricating, and adjusting of required parts.

Restorative Maintenance

0414. This includes testing and repairing of ICT equipment to restore them to their standard function, which ensures that the equipment can perform its continuous operations.

0415. In the event of an equipment breakdown, or facility failure, the following guidelines shall apply:

a. Conduct technical test procedures, as prescribed in user-level training for possible restoration of the C4S equipment and facilities, with the supervision of the designated signal officer (S6).

b. Upon confirmation of the C4S equipment breakdown and/or facility failure, individuals should send the said equipment to the concerned signal battalion of ASR placed operational control (OPCON) to the infantry divisions, for technical inspection.

c. An on-site technical inspection shall also, be carried out depending on the size, quantity of equipment, etc. at the request of the battalion/unit concerned to Division G6.

d. The signal battalion personnel can conduct technical inspections and evaluations on the unserviceable ICT equipment and facilities.

e. To provide a systematic guide on the steps to be performed for the repair of unserviceable equipment, the figures on the following pages are the process flow of repair conducted at The Headquarters ASR level (Figure 4.1); SIMBn Level (Figure 4.2); and C4S supplier level (Figure 4.3).

0416. Repair of C4S equipment still covered by service warranty and beyond the capability of SIMBn, ASR shall be sent to the concerned C4S equipment supplier to be processed as prescribed on the flow charts in the succeeding pages.

0417. Batteries of C4S equipment must be removed whenever they will be left for safekeeping at the designated storage area. If necessary, these batteries must be included in the packaging of unserviceable equipment for verification of its status and the possible cause behind its breakdown.

0418. No experiments, or use for the non-intended purpose, must be performed on C4S equipment. If there is a need to explore the use of C4S equipment, beyond the prescribed utilization, it must be conducted only by trained technicians of ASR.

0419. All C4S equipment must always be placed inside the issued carrying case when not in use.

0420. Unserviceable C4S equipment sent for repair must be enclosed in the issued carrying case with all the required accessories/ancillaries, as prescribed/required by ASR.

0421. Concerning units of ASR must not receive C4S equipment sent for repair not placed inside the issued carrying case and the required accessories/ancillaries.

0422. HPA may front load and preposition priority C4S equipment, accessories, spare parts, and other supplies to the appropriate PA unit, subject to its availability and on a need basis. These include items that are allocated for "Direct Exchange" to field units, subject to the existing logistical procedures.

0423. ASR shall establish a database for the repair of C4S equipment with data on, but not limited to: serial number, date issued, responsible PA unit, repair conducted, accessories/ancillaries, and spare parts replaced, among others. This database shall serve as a reference for future repairs of said equipment.

0424. ASR shall update the recipient PA units on the status of their respective C4S equipment sent for repair, through the headquarters of the PA major subordinate units.

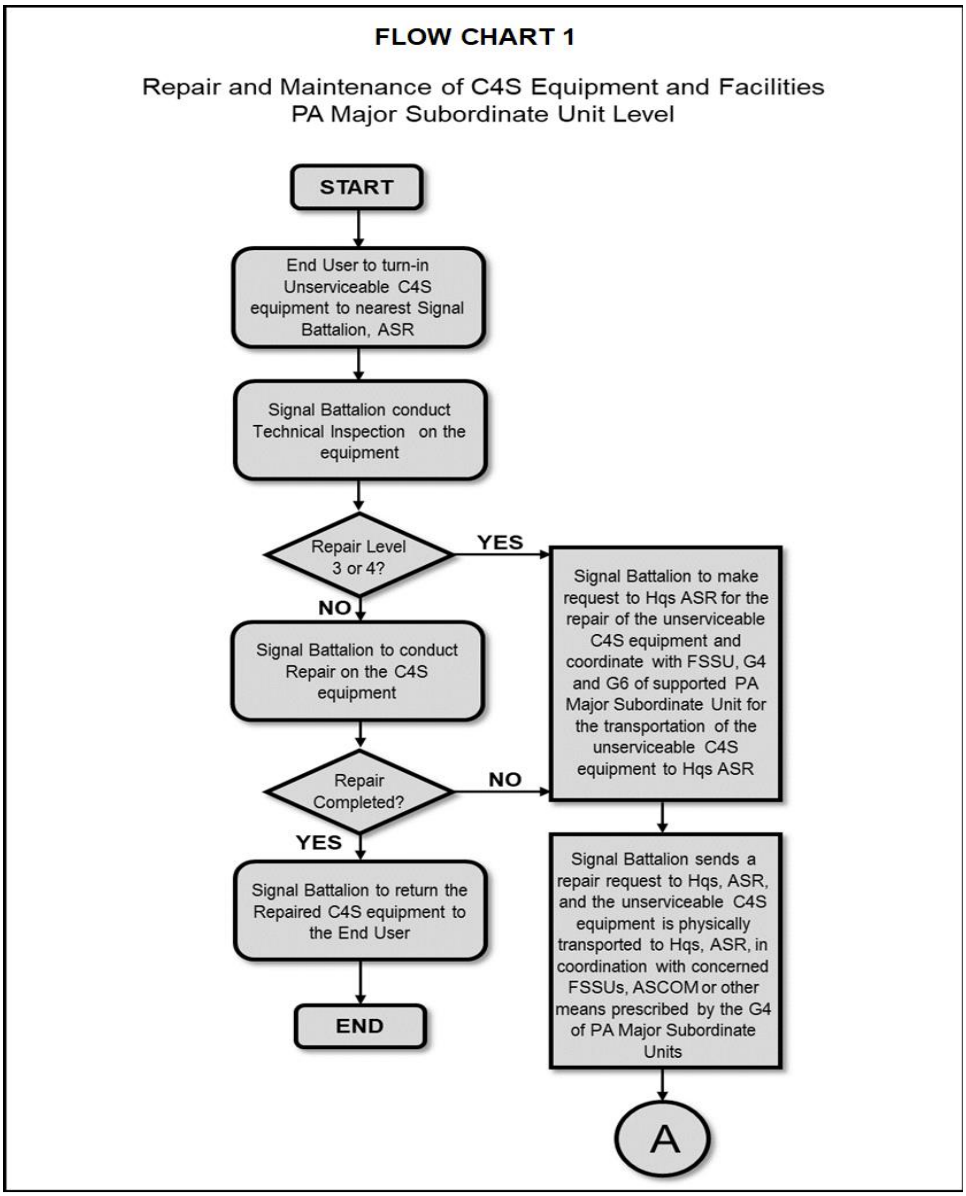


Figure 4.1 – Headquarters ASR Level Repair

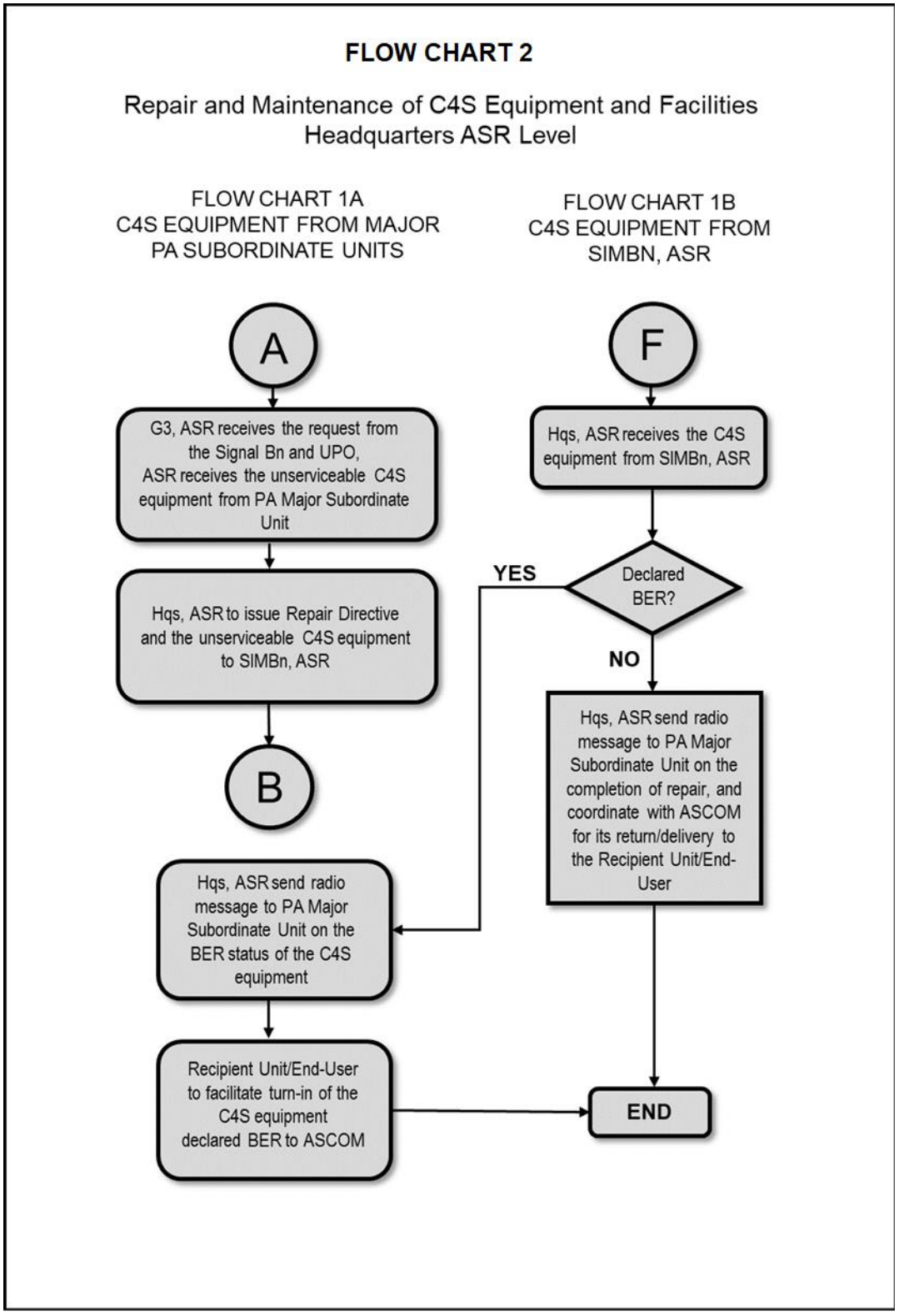


Figure 4.2 – SIMBn, ASR Level Repair

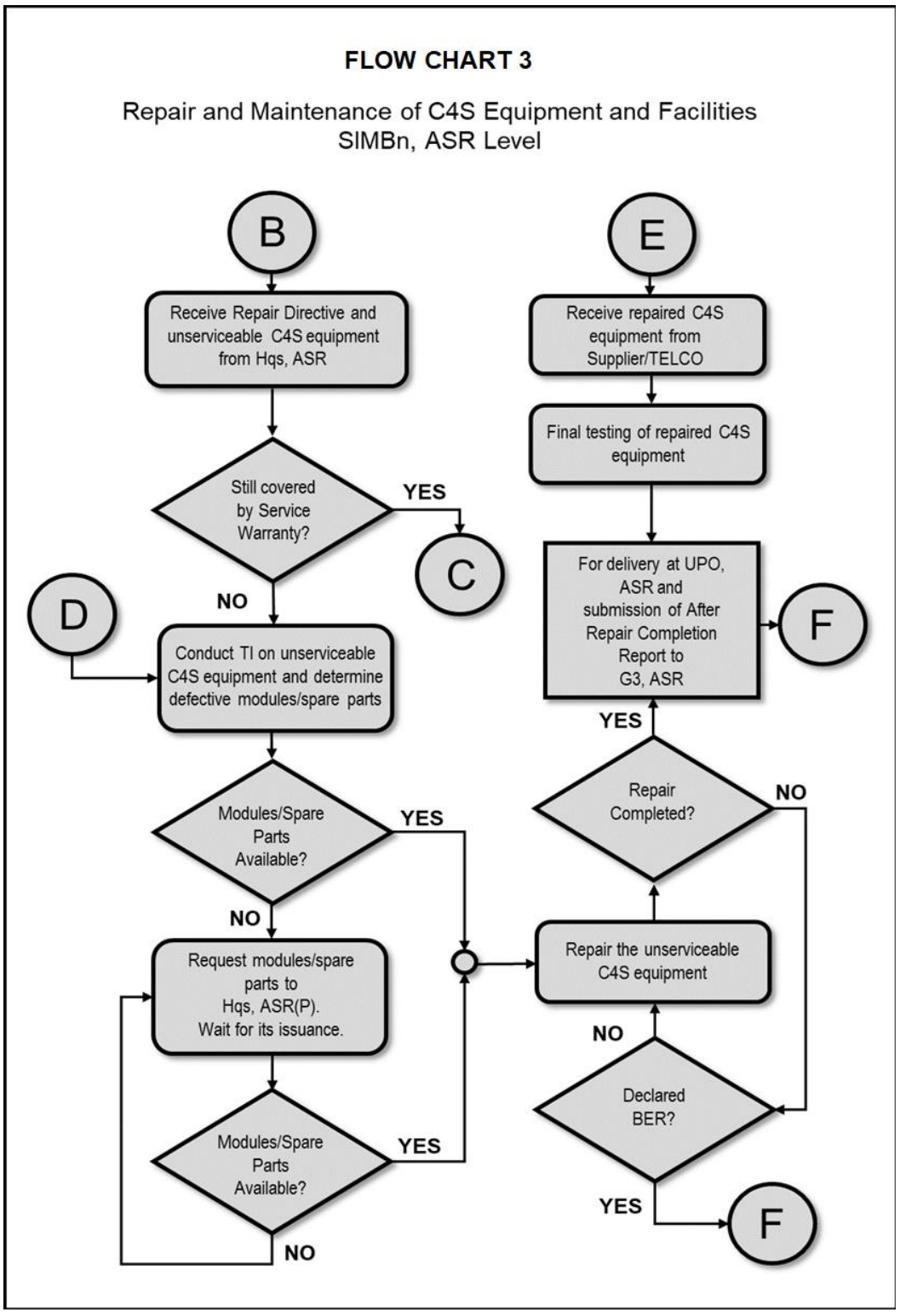


Figure 4.3 – C4S Supplier Level Repair

Disposal of Command & Control Communications and Cyber Systems Equipment

0425. The following are institutionalized standards for the disposal of C4S equipment to ensure that appropriate evaluation and approval processes are implemented.

0426. All C4S equipment declared beyond economic repair (BER) must be turned over to the nearest Forward Service Support Units, Army Support Command (FSSU, ASCOM) for proper disposal. The using unit must keep a copy of the property turn-in slip (PTIS).

0427. No spare parts, accessories, and ancillaries of C4S equipment must be disposed of by the using unit without proper authorization.

0428. For safety reasons on the potential hazard to humans and the environment, only FSSU, ASCOM shall dispose of turned-in ICT equipment, spare parts, accessories, and ancillaries that are declared.

Signal Sustainment

To ensure continued serviceability of signal equipment, maintenance facilities are established to cater to the repair requirements of supported units.

0429. Signal maintenance and training facility (SMTF) basic tasks:

a. Conduct Level 1 to 3 repairs of Tactical HF and VHF Radios

b. Conduct Level 1 to 2 repairs for UHF Commercial Radios

c. Programming of both Commercial and Tactical Radios

d. Equipped with an indoor classroom for the conduct of C4S related training

e. Has additional ancillary multipurpose rooms for other training and admin requirements.

0430. Signal mobile maintenance and repair team (SMMART) basic tasks:

a. Conduct Level 1-2 Repair for UHF Commercial Radios

b. Programming of both Commercial and Tactical Radios

c. Conduct C4S related refresher training

0431. DND-AFP electro-optic repair and maintenance facility basic tasks:

a. Conduct Level 1 to 3 repairs of NFS equipment.

b. Serves as a training center for NFS repair and maintenance students.

Programming of Funds

0432. Programming for the repair and maintenance of C4S equipment and facilities on MOOE must be under the current command guidelines on the program-budget development guidelines (PBDG).

0433. HPA program directors shall program funds for the C4S enhancement deployed to PA subordinate units.

0434. PAMUs shall program funds for the maintenance of C4S equipment at their respective headquarters and subordinate units.

0435. Fund requirements for repair and maintenance beyond the capabilities of PAMUs must be requested to HPA through AC of S for C4S, G6, PA with the necessary data for evaluation.

Personnel Management

0436. The personnel management of signal corps members in the PA evolves through its four management stages:

a. Recruitment

b. Education and Training

c. Employment

d. Promotion

Recruitment

0437. The recruitment of applicants for signal corps is based on the courses they have acquired from tertiary schools and technical institutions with industry-standard certifications for both enlisted personnel and civilian employees. These certifications are not limited to information technology and communications.

0438. Initially, the applicants are required to undergo a series of written examinations and trade tests before gaining admission to a military training institution that conducts basic military training. Due to the limited number of skilled signal corps personnel the organization resorts to selecting applicants with exceptional skills related to information and communication technology through special enlistment that waives basic requirements for regular enlistment in the PA.

0439. The primary consideration in the screening of applicants is their acquired technical skills that may help in the operational aspect of communication network systems in the Army.

0440. For officers, the pool of signal corps officers is being sourced from new graduates of pre-entry or basic military training based on their acquired skills from

learning institutions and those who opted to join the signal corps and be awarded the signal corps Armed Forces of the Philippines Occupational Specialty (AFPOS).

0441. One of the strategic aims of the ATR is to recruit the best and brightest among the candidates joining the PA. With this initiative, the PA emphasizes the importance of recruitment and selection of applicants during the pre-entry level.

0442. Likewise, the signal corps recruits applicants based on skills they have acquired from technical institutions with industry-standard certifications for both enlisted personnel and civilian employees. These certifications are not limited to information technology and communications.

0443. Due to the required signal skills' highly technical nature, the PA also offers options, such as special recruitment, to attract qualified and skilled individuals to join the signal corps.

Education and Training

0444. The education and training of the PA are based on a vision to create "a highly professional and well-trained PA that will win the nation's wars and contribute to national development."

0445. To achieve this vision, the PA's training programs shall extensively prepare soldiers, leaders, and units to carry out various military missions and succeed through responsive and effective training.

0446. Experience at field training exercises, warfighting exercises, and/or communication-electronics exercises (COMELEX) shows the importance of allowing enough time in the training for signal personnel on C4 systems.

0447. The individual and team tasks to operate and maintain these systems are complex, and the skills are perishable. Signal personnel with regular training always keep them technically and tactically proficient.

0448. This is especially true for C4 systems not operated by signal personnel since typical radio operators who are less familiar with signal equipment and concepts need regular refresher training with these systems.

0449. To carry out its mission in providing signal support operations, the unit must undergo specialized training and skill upgrades for its personnel.

0450. The following illustration shows the specialized training programs and skills required for the signal support workforce.

0451. Recognizing the signal support workforce career program, the signal personnel must undertake leadership and skills development training to the following PA's specialized training institutions:

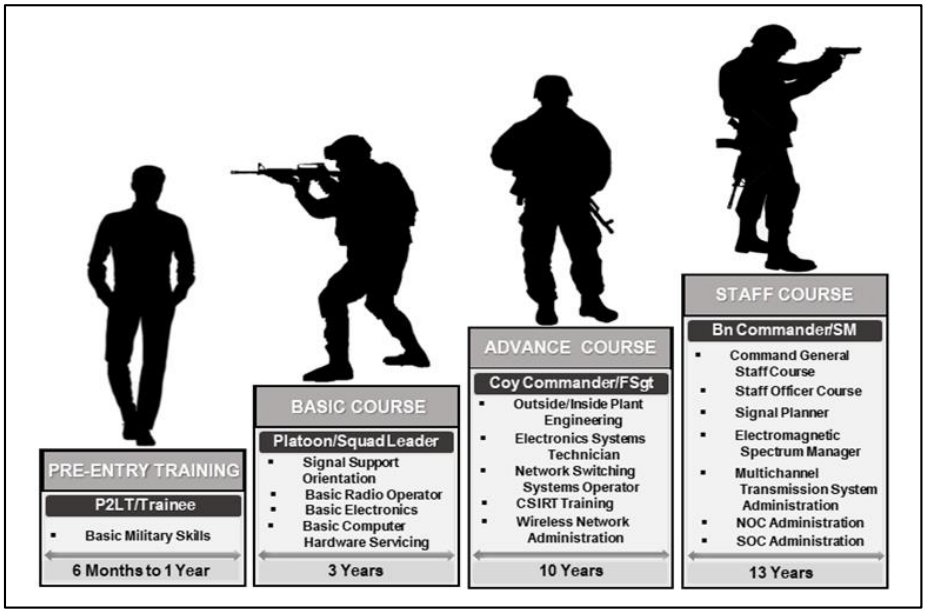


Figure 4.4 – Signal Training Path

Employment

0452. The assignment of signal personnel to Army units is dependent on the personnel requirements of the organization in consideration of the signal positions to be filled up at the strategic, operational, and tactical levels. Such function falls under the function of the Army Personnel Management Center (APMC), an office that implements the placement of signal-skilled personnel through the signal corps career adviser.

0453. Said office based its employment of signal personnel on the organization's mission-oriented policies, unit readiness conditions, and personnel skills and competencies assigned to a position.

0454. Part of its objective is to sustain a strong, mission-oriented career management program that shall direct the tasks and employment decisions.

0455. Signal personnel, whether officers, enlisted personnel, or civilian employees, should be equipped with the appropriate knowledge, skills, and attitudes necessary to carry out their assigned duties effectively.

0456. To determine the skill-based program of the signal personnel, a Signal Corps (SC) AFPOS is assigned.

Promotion

0457. Considered as one of the important factors in the career management of ASC personnel, promotion provides them the opportunity to hold higher position and responsibilities as they advance to the next higher rank.

0458. Among the personnel requirements for promotion to the next higher grade include the following: having satisfied the time in grade; holding positions commensurate to his/her rank; and Personnel Evaluation Report (PER).

Annex A
Signal Estimate Format

I MISSION
II SITUATION AND COA
AO and AI Analysis
A. TERRAIN:
B. WEATHER:
C. EXISTING COMMUNICATIONS:
D. ENEMY:
a. Organizational Structure:
b. Disposition and Location:
c. Strengths and Weaknesses:
d. Capabilities and Limitation:
E. OWN SITUATION:
a. Organizational Structure of Forces:
b. Disposition and Location of Friendly Forces:
c. Status of Personnel and Training Profile (ASSUMED):
d. Inventory of Signal Assets (ASSUMED):
e. Capabilities:
F. COURSES OF ACTION:
III ANALYSIS OF OPPOSING COA
A. COMMUNICATION AND DISPOSITION OF OPPOSING FORCES:
B. SIGNAL PROPAGATION OF OPPOSING FORCES:
C. LIST OF ENEMY CAPABILITIES AND COUNTERMEASURE (RFI from Intel):
Capabilities:
Limitations:
D. LIST OF FRIENDLY FORCES CABABILITIES AND LIMITATION:
Capabilities:
Limitations:
E. COUNTER-MEASURES:
On the Enemy Capabilities:
Signal Platoon Limitations:
IV COMPARISON OF COAs
V DECISION/RECOMMENDATIONS:
A. PERSONNEL REQUIREMENTS:
RECOMMENDATIONS:
B. C4S REQUIREMENTS:
RECOMMENDATIONS:
C. SUSTAINMENT REQUIREMENTS:
RECOMMENDATION:
D. MEANS OF COMMUNICATION:
E. LOCATION OF TCP AND MCP:
F. TASK LIST (Implied and Specific) (CMDR’S DECISION)

Annex B
Signal Plan Format

Copy Nr _____ of _____ Copies
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SIGNAL PLAN "HARBINGER" to OPORD 01-2020 dtd 05 April 2020

References:

Time Zone Used Throughout the Plan: HOTEL

Task Organization:

1. SITUATION

a) Enemy Forces:
b) Friendly Forces:
c) Attachments and Detachments:
d) Facts

I. ENEMY FORCES

II. FRIENDLY FORCES

2. MISSION

3. EXECUTION

a. Concept of Operations

b. Concept of Support by Phase

PHASE 1 – PLANNING AND PREPARATION

Task:
Purpose:
Method:
Effect:

PHASE 2 EXECUTION PHASE

Task:

Purpose:
Method:
Effect:

PHASE 3 CONSOLIDATION AND REORGANIZATION

Task:
Purpose:
Method:
Effect:

c. Task to Manuever Unit.

d. Task to Support Unit.

1) Signal Section
2) Intel Section
3) Operations Section/
4) Logistics Section/TF24:

e. C-E Support:

f. Coordinating Instructions:

1) Primary means of communication is Harris Handheld VHF Tactical radio, alternate is Harris MP Tactical radio and emergency is a messenger.

4. SUSTAINMENT:(See OPORD)

5. COMMAND AND SIGNAL

a. Command.

b. Signal.

Acknowledge:

LTC

Official:

Glossary

Command and Control, Communications, and Cyber Systems
The system vital for command, control planning and guidance, and in the dissemination of information. It also performs logistics, intelligence, and operation functions. C4S serves as a central system that integrates all aspects of operations and allows the effective and efficient provision of command and control communications.

Command and Control Communications System
The organization of resources that the commander uses to help him plan, direct, coordinate and control military operations to ensure mission accomplishment.

Command and Control System
The arrangement of personnel, processes and procedures, networks, and command posts that enable commanders to conduct operations.

Command and Control Warfighting Function
The related tasks and a system that enable commanders to synchronize and converge all elements of combat power.

Command and Control
The exercise of authority and direction by a properly designated commander over assigned and attached forces in the accomplishment of a mission.

Command
It is the authority that a commander in the military service lawfully exercises over subordinates, by virtue of rank or assignment.

Communications
A method or means of conveying information of any kind from one person or place to another.

Cyberspace
A global domain within the information environment consisting of the interdependent networks of information technology infrastructures and resident data, including the Internet, telecommunications networks, computer systems, and embedded processors and controllers.

Information Environment
The aggregate of individuals, organizations, and systems that collect, process, disseminate, or act on information.

Information System
A computer-based sourcing, manipulating, and storing data for the purpose of generating information for the use of organizations.

Information Technology
Refers to computers, telecommunications, and office systems technologies.

Mission Command
The shared understanding between commanders and his subordinates of the environment and its expectations which enable every member to exercise disciplined initiative.

Radio Net
It is composed of two or more radios operating in the same frequency for the specific purpose of interchanging information.

Signal Corps
The service branch of the PA primarily tasked in the management and operations of Army C4S.

Signal
Signal is a means of transmitting information with corresponding pre-arranged system or codes in cases where direct verbal or written statement is unnecessary, undesirable, or impractical. It is the collective and integrated use of Command and Control Communications, and Cyber Systems (C4S) to support warfighting capabilities across all levels of command. Signal support operations are highly essential for the successful execution of Army operations.

Signal Operations
Signal operations is a synchronized action of signal personnel and units at all echelons providing and securing a network for Army commanders to conduct command and control, and integrating other warfighting functions across a range of Army operations.

Abbreviation and Acronym list

ADO	Active Defense Operations
AFP	Armed Forces of the Philippines
AFPOS	Armed Forces of the Philippines Occupational Specialty
APMC	Army Personnel Management Center
ASC	Army Signal Corps
ASR	Army Signal Regiment
ATR	Army Transformation Roadmap
BER	Beyond Economic Repair
BONTEX	Bonifacio Telephone Exchange
C2	Command and Control
C2C	Command and Control Center
C4S	Command & Control Communications and Cyber Systems
CEIS	Communication Electronics and Information Systems
CEOI	Communication-Electronics Operating Instructions
CESI	Communication-Electronics Standing Instructions
CMO	Civil-Military Operations
CNR	Combat Net Radio
CO	Capital Outlay
COMELEX	Communication-Electronics Exercises
COMMEL	Communications Electronic
COMSEC	Communications Security
CSBn	Command Signal Battalion
CSIRT	Computer Security Incident Response Team
CSR	Connect, Sustain and Recover
CUP	Capability Upgrade Program
CyBn	Cyber Battalion
DBTOCS	Deployable Brigade Tactical Operations Center System
DCO	Defensive Cyber Operations
DND	Department of National Defense
DST	Deployable Signal Teams
EM	Electromagnetic
EMS	Electromagnetic spectrum
EW	Electronic Warfare
FCS	Fixed Communication Systems
FSMMART	Forward Signal Mobile Maintenance and Repair Team
GHQ	General Headquarters
HADR	Humanitarian Assistance and Disaster Response
HF	High Frequency
HPA	Headquarters Philippine Army
ICS	Incident Command System
ICT	Information and Communications Technology
INO	Information Network Operations
IRP	Incident Response Plan
ISR	Intelligence Surveillance and Reconnaissance
MCC	Mobile Command Center
MDMP	Military Decision-Making Process
MNLF	Moro National Liberation Front
MOOE	Maintenance and Other Operating Expenditures

MOUT	Military Operation in Urban Terrain
NDRRM	National Disaster Response and Risk Mitigation
NETBn	Network Enterprise Technology Battalion
NFS	Night Fighting System
NTC	National Telecommunications Commission
OE	Operational Environment
OPCON	Operational Control
OTH	On-The-Halt
OTM	On-The-Move
PA	Philippine Army
PAM	Philippine Army Manual
PAMU	Philippine Army Major Unit
PANET	Philippine Army Network
PAS	Public Address System Support
PASO	Philippine Army Signal Operations
PBDG	Program-Budget Development Guidelines
PER	Personnel Evaluation Report
ROIP	Radio over Internet Protocol
SBn	Signal Battalion
SC	Signal Corps
SIMBn	Signal Installation and Management Battalion
SIMCoy	Signal Installation and Management Company
SMMART	Signal Mobile Maintenance and Repair Team
SMTF	Signal Maintenance and Training Facility
SO	Signal Operations
SOP	Standard Operating Procedure
SPC	Special Purpose Communication
SSO	Signal Support Operations
TCP	Tactical Command Post
TE	Table of Equipment
TLP	Troop Leading Procedure
TO	Table of Organization
TOC	Tactical Operations Center
TSS	The Signal School
UHF	Ultra High Frequency
VHF	Very High Frequency

References

Army, Department of the. (2019). *FM 6-02 Signal Support to Operations*. *13 September 2019*.

Army, Department of the. (2020). *TECHNIQUES FOR TACTICAL RADIO OPERATIONS Headquarters , Department of the Army This publication is available at the Army Publishing Directorate*. *January 2016*.

Army, U. S. (2014). *Optimizing C4ISR field support for today's Army*. https://beta.genei.io/projects/demo-faf5106a-2303-41b0-a23e-e311be7fd56b/2097c1d0-fefb-4f29-aeef-59aa5f590548

Army Transformation Road Map

Blumberg, M. S. (2020). *The Integrated Tactical Network Pivoting Back to Communications Superiority*. *6*(June).

Frater, M. R., & Ryan, M. J. (2015). *A MODERN TACTICAL COMMUNICATIONS ARCHITECTURE FOR THE AUSTRALIAN ARMY*. *September 2000*.

Joint Staff, U. (2018). *Joint Publication 3-12 Cyberspace Operations*. *June 2018*.

Matthews, T. (2018). *Complete Guide to CSIRT: How to Build an Incident Response Team*. Exabeam. https://www.exabeam.com/incident-response/csirt/

Philippine Army, P. (2007). *PAM 6-00, COMMAND AND CONTROL COMMUNICATIONS , AND CYBER SYSTEMS*. *23 October 2007*.

Philippine Army, P. (2020). *PAM 6-0102-04, Cyberspace Operation*. *February 2020*.

Thales. (n.d.). *Electronic Warfare*. Retrieved June 17, 2021, from https://beta.genei.io/projects/demo-faf5106a-2303-41b0-a23e-e311be7fd56b/9740330f-ff3a-4dc4-a4c7-72db762ab0f7?tab=summary

Victoria, A., & Christianity, F. (2019). *Electronic Warfare*. *October 2019*.

PA Operating Concept 2021

PA C4S Operational Thrust and Guidelines

National Security Policy 2017-2022